ECE 551 Design Vision Tutorial

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Lesson 0 – Tutorial Setup

NOTE: Throughout the tutorial, there are commands (denoted by `%>`) you need to enter into the console/terminal. **Do not directly copy** them from the document, as some characters might not encode correctly in Unix. Type everything (that means everything but the leading `%>`) manually instead.

DEFINITIONS:
Synopsis: Company that makes the tool
Design Compiler (DC): The synthesis tool
Design Vision (DV): A GUI for interacting with DC

1. From your home directory, go to your ece551 directory (as should have been created in the ModelSim tutorial. Otherwise create a directory named "ece551" and then switch to it).

%> cd ece551

2. Copy all files from the Design Vision tutorial directory to your current directory

%> cp -r ~ejhoffman/ece551/public/designvision ./

3. Now that everything is copied, change the directory to the tutorial directory

%> cd designvision

Choosing Your Tech Library:

`.synopsys_dc.setup` is a configuration file that allows you to choose between multiple different standard cell libraries, or even various options with a standard cell library.

The tutorial directory contains a ready-made **`.synopsys_dc.setup`** file. It utilizes the Synopsys 32nm educational library (***saed32***).

NOTE: Did you use the **`%> ls`** command to check for the file but couldn't find it? Turns out that files starting with a '.' are hidden by default in Unix (the current operating system). To display them, use **`%> ls -a`** instead. ☺

Choosing your PVT
IC manufacturing processes often come with variations that affect the speed of the transistors, thus the timing of the gates.

For instance, the process could be run with implant level and gate oxide thicknesses that result in low threshold voltages (low Vt). This would result in a faster process, but higher power because transistors would leak more. The voltage the design runs at also affects transistor/gate speeds.

Higher voltage is faster but more power hungry than lower voltage. Finally, temperature also affects transistor speed. Cold is faster, hot is slower. These 3 factors (**P**rocess parameters, **V**oltage, **T**emperature) are referred to as a "PVT corner".

We are going to choose a "middle of the road" PVT corner. Thus, before continuing, let's confirm that the provided **`.synopsys_dc.setup`** file chooses the proper PVT corner.

4. Using any text editor of your choice (***gedit*** might be a good choice), open the **`.synopsys_dc.setup`** in the current directory

`%> gedit .synopsys_dc.setup`

Near the top of the file you will see various set commands for the PVT corner parameters. Some are commented out (# comments out everything to the right until the end of the line). We want to ensure the following are being used (i.e uncomment them if they are commented while also ensuring the ones that we don't want are commented).

```
set VT rvt              # rvt = regular Vt
set PROCESS tt          # tt => nmos typical, pmos typical
set VOLT med            # med = medium voltage
set TEMP room           # room temperature
```

5. Save the .synosys_dc.setup file (**`ctrl + s`**, **`cmd + s`**, or the "Save" button in the upper right corner) and exit the text editor.

6. Copy the file from the current directory to your home directory.

%>cp .synopsys_dc.setup ~/

NOTE: When you launch Design Vision (as we will do in a moment), the tool will first check your current directory for a valid **`synopsys_dc.setup`** file. If it cannot find one there, it then checks your home directory. Thus, always be aware of location of the configuration file you care about is (if you need to alter it and/or launch Design Vision).

7. Change to the main tutorial directory

%>cd dv_tutorial

It is important to start Design Vision from the directory where your project (Verilog code, constraints file, synthesis script – optional, configuration file – if not using one in the home directory) is so that environment variables that are automatically setup will be made correctly.

IMPORTANT
If at any point in the tutorial, you see that the design variables (values of VT, PROCESS, library, etc.) do not match what we established above:

*In the directory you started Design Vision in remove the **`.synopsys_dc.setup`** file.

`%> rm .synopsys_dc.setup`

Then launch Design Vision again. This will force it to use the one from your home directory.

8. Start Design Vision in GUI (Graphical User Interface) mode

%> /cae/apps/bin/design_vision

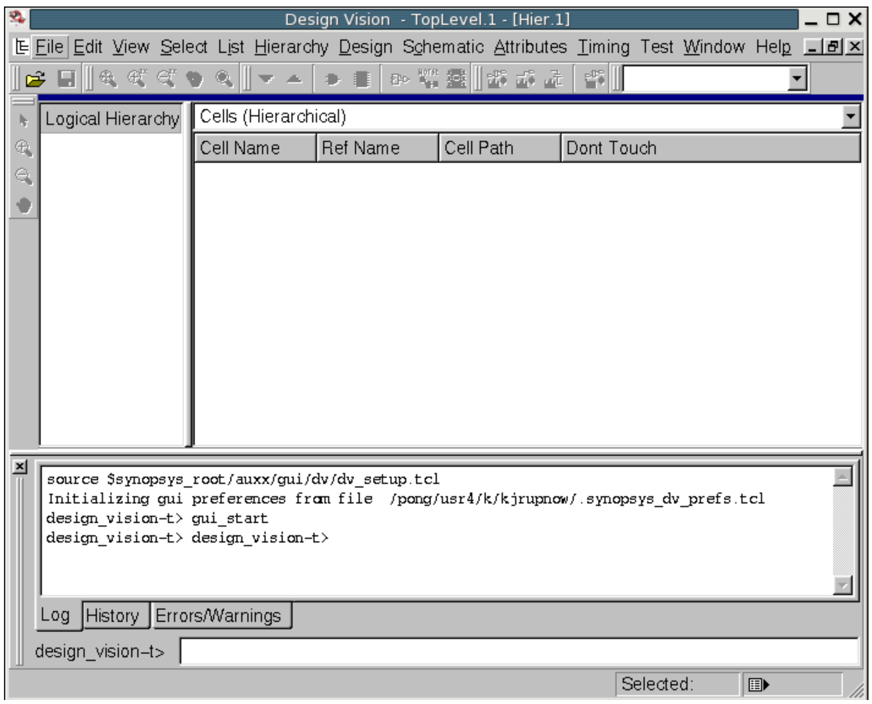


Figure 0-1 : Design Vision Default Window

We may look at the main window of Design Vision to see a few of the program's features.

The left pane is for the **"Logical Hierarchy**". Once a design has been selected, it will show its entire hierarchy. We haven't selected a design yet, so it should be empty.

The right pane is the **"Cells (Hierarchical)"** window. It will show a bit more information about the hierarchy depending on which level (of the hierarchy) is selected on the left. We haven't selected a design yet, so we do not have a hierarchy to inspect (i.e it should be empty).

The bottom panel has two different tabs: **log** and **history**. The important thing to note about this panel is that every command you perform will populate each of these tabs with some information.

The log tab will keep spit out information during the synthesis process (including errors and warnings. In particular, if your compilation process [more on what that means later] ends in a 0, there was an error. If it ends in a 1, it was a success. All of this is in the log window).

Additionally, the history tab is particularly useful, as you can inspect the contents and learn synthesis commands to make your own scripts later.

Lesson 1 – Code Input (Reading Verilog files via DV)

1. File -> Read

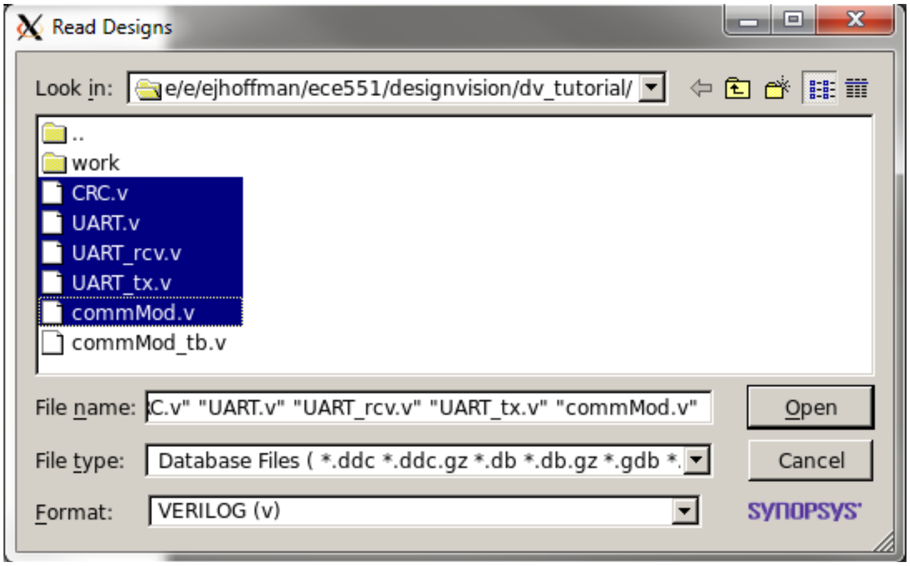


Figure 1-1 : Reading Verilog Files

Select all the Verilog files except commMod_tb.v. This module is the testbench, so is not synthesized. Set the format to be Verilog, or SVERILOG (if using system Verilog). Click Open

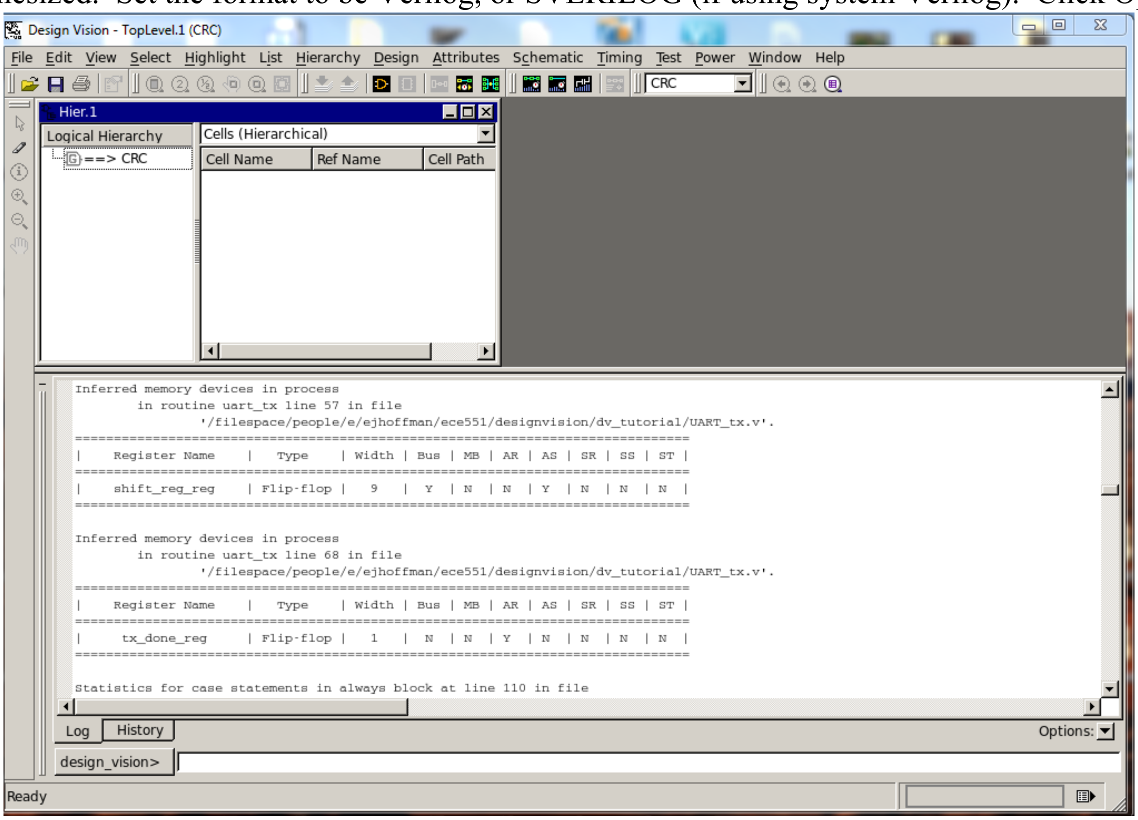


Figure 1-2 : Log Results of Verilog Read

If you expand the bottom panel and look at the **log** tab, you should see information pertaining to the Verilog files Design Vision just read. Specifically, you can see that by reading the files it has inferred several different flops. This is always a good section to look at and ensure that the flops inferred by Design Vision are flops you intended to make when writing your Verilog code (i.e make sure that there are no latches mistakenly inferred in your design).

If you now look at the **history** tab, you should see a **`read_file`** command. This is an example of a command you can learn and use to make scripts later for other designs (*cough* the final project *cough*).

2. Set Current Design to top level (commMod)

When we read files in, we need to ensure that the synthesis tool knows which Verilog file is our top level design (i.e the top of the hierarchy). For this tutorial, the top level design is a module called commMod and we need to make sure that it is set as such.
First, let's list all of our files.

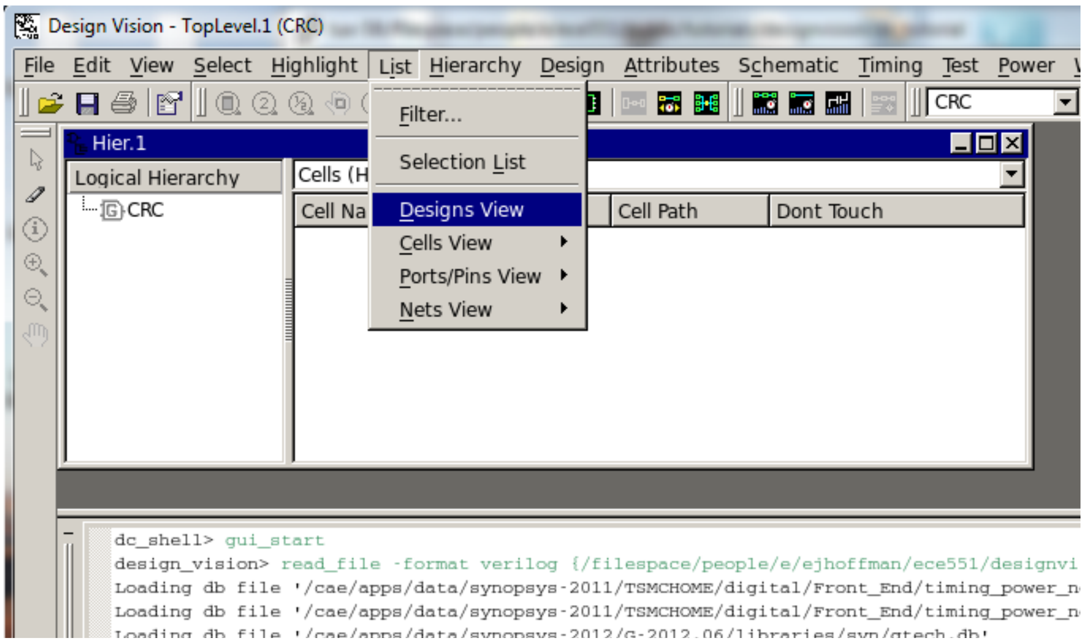


Figure 1-3 : Set Current Design (part I)

Now we set commMod to be the current design we are working on (top level). Click on commMod to select it then right click and choose “Set Current Design” from the context menu.

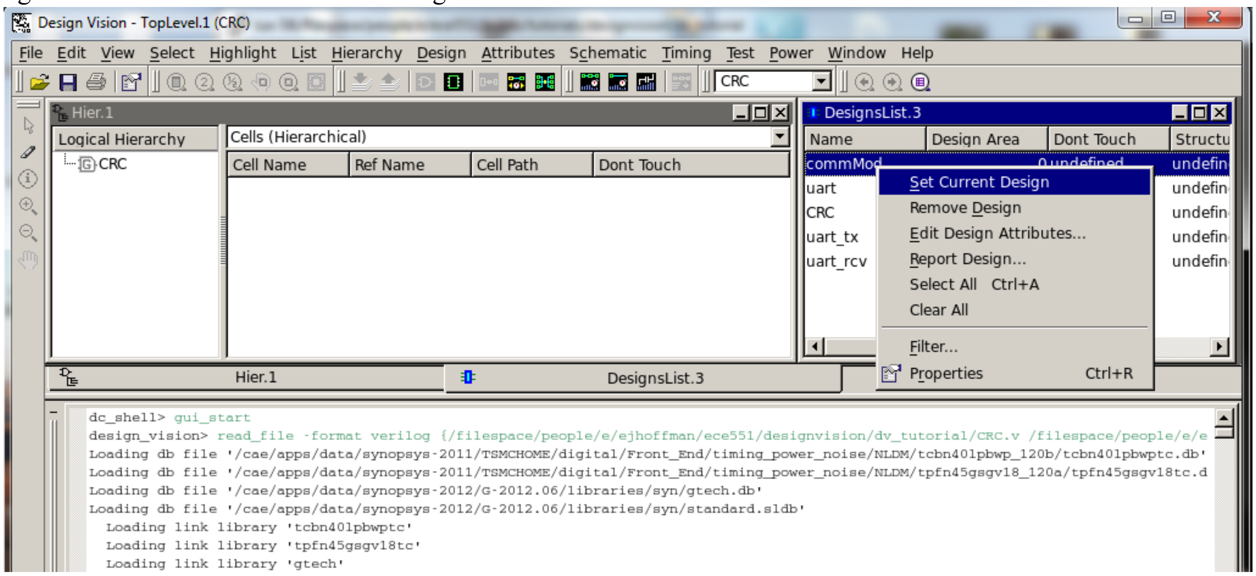


Figure 1-4 : Set Current Design (part II)

3. Viewing Pre-Synthesis Schematic

Once in a while it can be useful to view the schematic view of what Synopsys thinks the circuit is. These schematic views are auto generated and often hard to follow. Still if it is your design (you wrote the Verilog and are familiar with it) then sometimes it will make sense to you and give you some insight into what Synopsys is “thinking”.

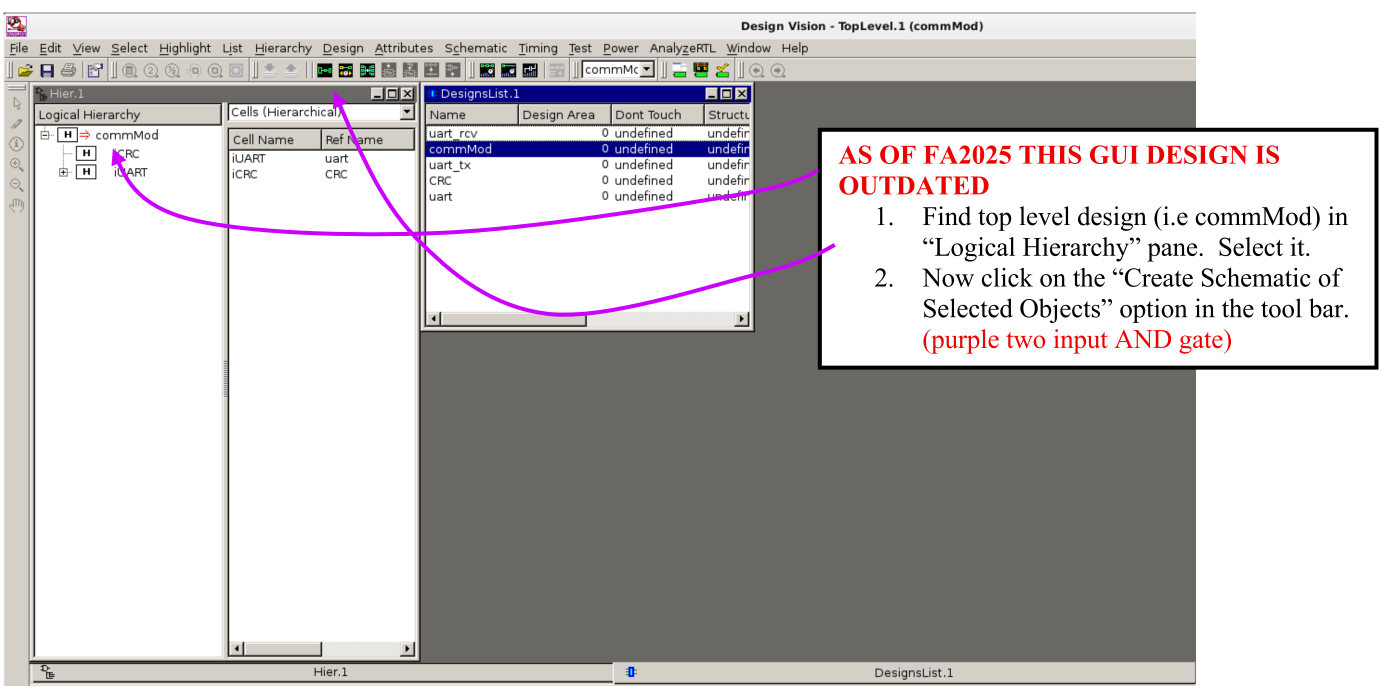


Figure 1-5 : Schematic Generation

Once the schematic is generated you can view (double click on elements to go down a level of abstraction, right click and select "back" to go up a level). Play around a bit with this. At this point we have not actually synthesized the design. We have read it in, and analyze and elaborate (i.e a fully connected RTL-level netlist) has been performed. The design is not yet mapped to our standard cell library, but rather just mapped to Synopsys GTECH cells (generic functional blocks).

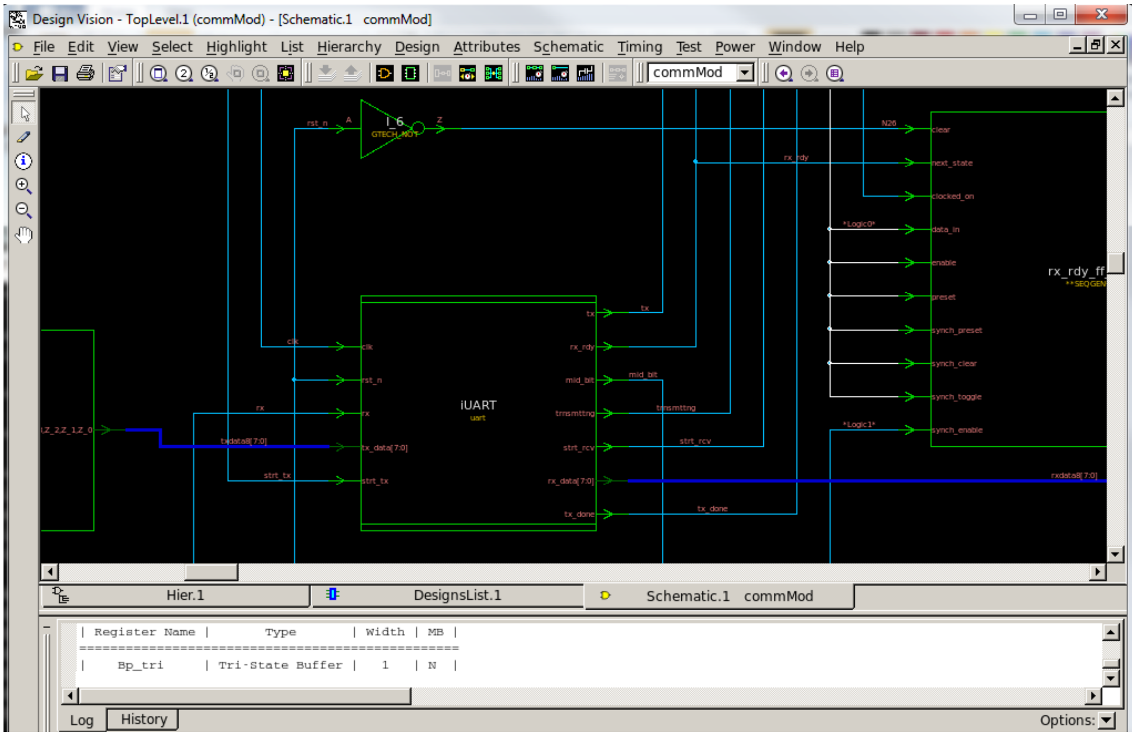


Figure 1-6 : Viewing a Schematic

Lesson 2 - Simple Synthesis (with no constraints)

Now that we have analyzed and elaborated the design, we can perform simple synthesis on our design. In ECE 551, we will require the use of design constraints which will be covered in Lesson 3. However we may see some information about the design by performing an unconstrained compilation, so we will do that here.

So now that we have an elaborated design (a generic RTL netlist), we need to map it into actual standard cells from our selected library (i.e make the hardware). This process also involves optimizing the design to meet certain goals (like timing, area, and power). The process of performing mapping and optimization together is called "compiling". The behavior of the synthesis tool during compilation can be influenced by setting options related to compilation and/or optimization.

The optimization part is specified by setting "optimization goals" such as:

1. **Mapping Effort** – how thoroughly the synthesizer searches for the best mapping of RTL logic to standard cells.
2. **Area Effort** – how aggressively the synthesizer tries to minimize total circuit area.
3. **Power Effort** – how aggressively the synthesizer tries to reduce power consumption.

Compile options include:

1. Top Level – compiles only the top level of the design, leaving the rest uncompiled
2. Ungroup All – Ungroups the entire hierarchy so that all logic is compiled as one module
3. Scan – Refers to the insertion of scan chains for testing logic. This should not be used for this course.
4. Incremental Mapping – specifies that the mapping should be done based upon only local information instead of global information. Furthermore, if some mapping information is already available, the synthesis will start from the previous map. This is especially useful when trying to improve upon an initial synthesis
5. Allow Boundary Conditions – allows that boundary conditions such as known input constants can be used to help optimize the design
6. Auto Ungroup – when enabled, you may choose either area or delay as the trigger, and the synthesizer will automatically ungroup designs to meet constraints if the constraints are not being met for the trigger

Okay… enough talking. Let's compile. Navigate to the top menu bar.

Design -> Compile Design

*Leave options as is, just hit okay.

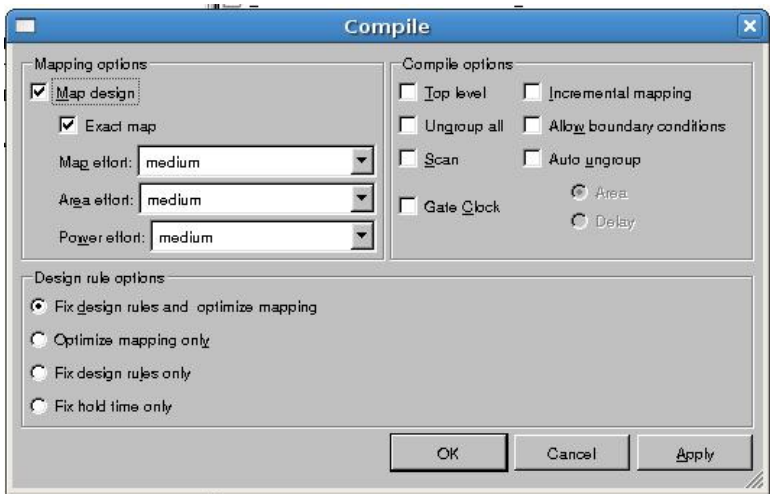


Figure 2-1 : Compile Design Dialog

The "**Compile Ultra"** command uses the same compile options as "Compile" command (what we just ran), which have the same meaning as previously described. The difference is that all effort options are automatically set at their highest levels and the synthesizer automatically picks options to achieve the best results. Compile Ultra should be used carefully as it will always take longer time than manually selecting the options (performs multiple optimization passes, etc.).

As compilaiton occurs, you can see the log window in Desgin Vision quickly populating with a lot of text. One of the things it's outputting during this stage are tables that specify where it is currently at with regard to minimizing the synthesis cost function (how small and fast can it make our design while still meeting design rules?).

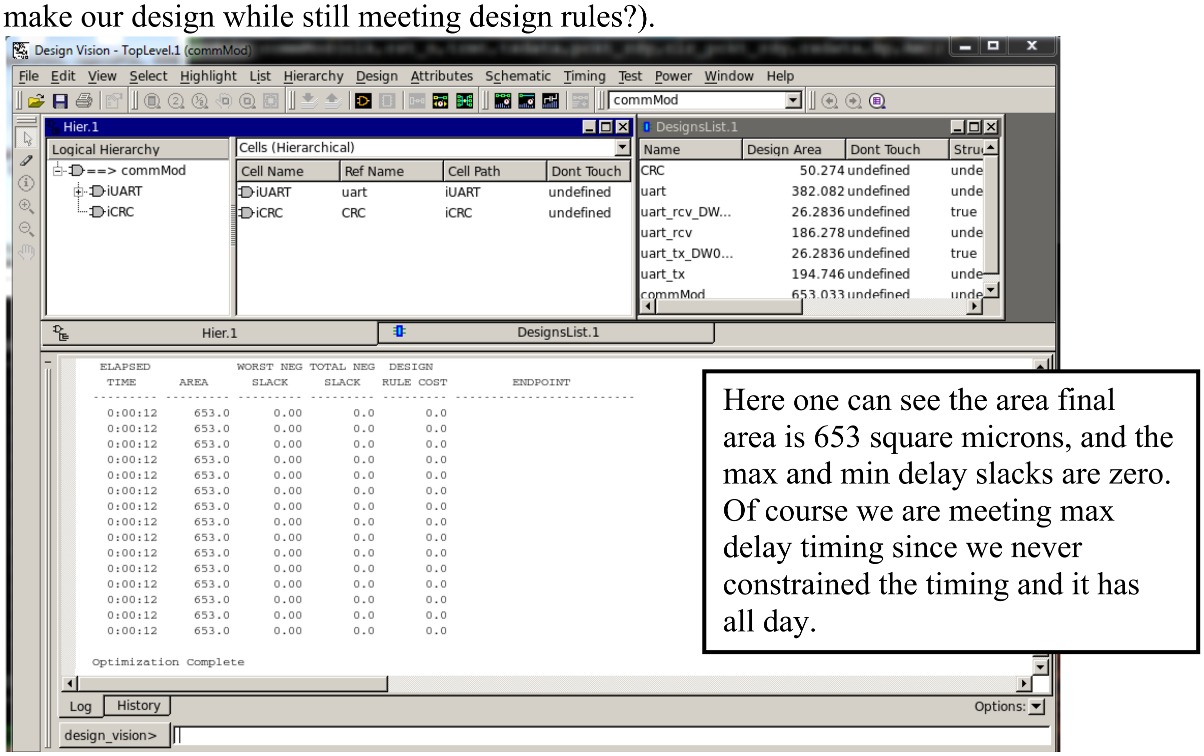


Figure 2-2 : Compile Results in Log Area

Lesson 3 – GUI's Are For Children

OK, I'm not saying that all GUI's are bad. In many CAD tools the GUI interface more intuitive one than the command based one. However, with Synopsys Design Compiler/Design Vision this is not the case.

Design Compiler was originally a command line tool, and the GUI (Design Vision) was bolted around it in later years… and it shows. This tool is much better run in command line mode. If you run it command line you can seamlessly develop a script with all of your synthesis commands. This facilitates reproducibility by giving you something you can easily re-run later.

Quit the GUI (hit the "X" at the top right… you might have to enter **`ctrl + c`** a few times in the terminal as well).

While still being in terminal, make sure you are in the same directory as the Verilog files.

1. Invoking Design Compiler in shell mode

```
%>design_vision -shell dc_shell
```

Now the wonderful **D**esign **C**ompiler shell (dc_shell) should be up with a prompt like: "**`dc_shell`**>"

2. Reading in the Verilog

```
dc_shell> read_file -format verilog { UART_tx.v UART_rcv.v UART.v \
CRC.v commMod.v }
```

Does the above command look familiar? (*cough* history window in lesson 1 *cough*)

NOTE: The "\" at the end of the line means the command continues on the next line. Lists of design files are included in {}.

3. Setting Current Design to Top level (to commMod)

```
dc_shell> set current_design commMod
```

4. Compiling (actually synthesizing)

```
dc_shell> compile -map_effort medium
```

Now we are already back to where we were when using the GUI, and it was actually much more direct to do it with the command line interface.

Now we will write out our resulting gate level netlist as a .vg file.

`dc_shell> write -format verilog commMod -output commMod.vg`

Now you can quit dc_shell using the **`quit`** command and do a 'more' on **commMod.vg** to see the Verilog gate level netlist that was produced.

`%>more commMod.vg`

Note that the result is structural Verilog that instantiates standard cells from our library. Type q to quit browsing the netlist file.

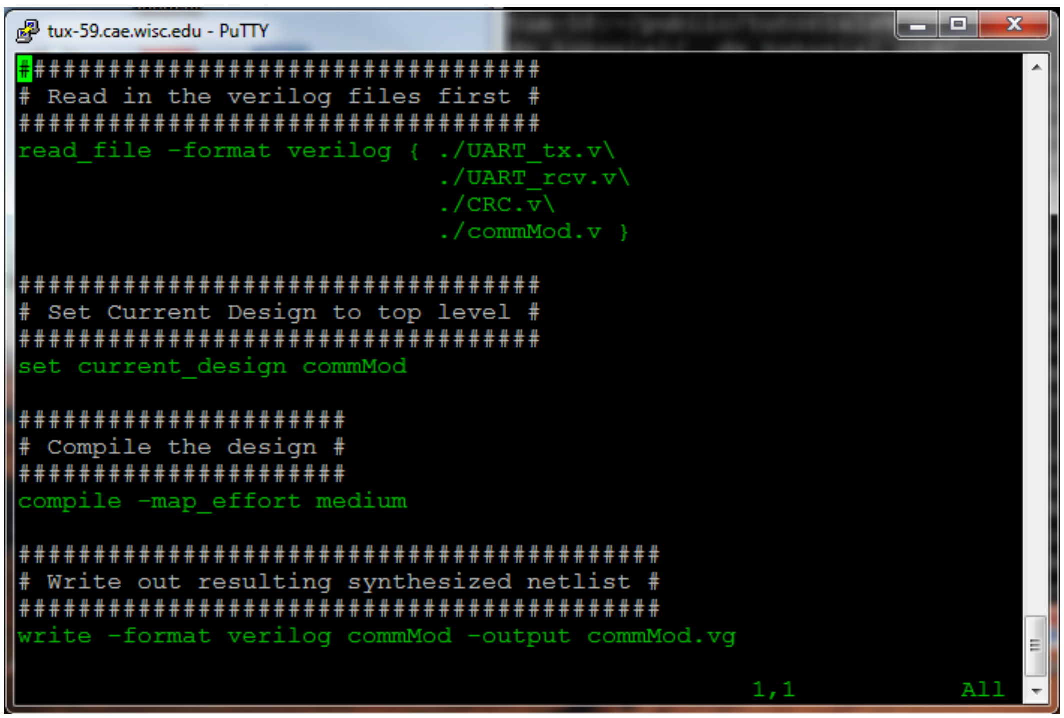
Lesson 4 – Synthesis Script (why not save my work)

It should be obvious that a command line driven CAD tool should be driven with a script as opposed to just typing the same commands over and over again. In this section we will develop a synthesis script.

A synthesis script is simply a text file with all the dc_shell command used. Of course as with any script/program some commenting is always a good idea.

1) Create Shell Script (preferably with a .dc file extension)

Using a text editor of your choice create a file called: **`commMod.dc`**




```
####################################
# Read in the verilog files first #
####################################
read_file -format verilog { ./UART_tx.v\
                            ./UART_rcv.v\
                            ./CRC.v\
                            ./commMod.v }

####################################
# Set Current Design to top level #
####################################
set current_design commMod

######################
# Compile the design #
######################
compile -map_effort medium

###########################################
# Write out resulting synthesized netlist #
###########################################
write -format verilog commMod -output commMod.vg

                                                  1,1           All
```

Figure 4-1 : Simple Synthesis Script

The above script is the same as what we did in Lesson 3. It is a simple synthesis script with no timing constraints.

2) Running a Synthesis Script

Method 1: Sourcing the script from within dc_shell

Kick off a dc_shell from a Unix terminal

```
%>design_vision -shell dc_shell
```

Now source your synthesis script from within dc_shell

```
dc_shell> source commMod.dc
```

Method 2: Invoking both dc_shell and the script from the Unix prompt

```
%>design_vision -shell dc_shell -f commMod.dc
```

Try one of these two methods.

NOTE: make sure you are still in the same directory as your project! (i.e the Verilog files and your synthesis script should all be in the same directory and this is where you should be running the script from)

Lesson 5 – Constraining The Design

NOTE: In this lesson, you will be making additional edits to your synthesis script (commMod.dc). **Do not** run these commands individually in dc_shell. While it's technically possible, that's not the focus of this lesson. Right now, we are concentrating on writing the script itself. Please continue reading, he instructions will tell you exactly what the new commands are and where to insert them.

The synthesis tool tries to optimize the design by minimizing its cost function. This cost is based on meeting design rules, meeting timing, and minimizing area & power. To have a properly synthesized design one needs to have a properly constrained design.

1) Constraining the clock

The clock frequency at which your design runs by is one of the most fundamental of timing constraints. Clock period specification, and the pin it is sourced to (signal/wire it is connected to in your top level design) is done as follows:

```
dc_shell> create_clock -name "clk" -period 2 -waveform {0 1} {clk}
```

The command above is creating a clock with period of 2 time units (nano seconds in our case). The waveform has 50% duty cycle since it starts high at 0ns, and falls at 1 time unit into the period. The clock has a name of "clk" (this is a handle). It is sourced to a pin **clk**.

The clock goes to a lot of flops in your design. It is a high fanout net (the heartbeat of your design). We don't want Synopsys doing anything stupid with it like buffering it. Clock distribution is typically handled in a separate tool. We want to tell Synopsys not to mess with it.

```
dc_shell> set_dont_touch_network [find port clk]
```

Add the above two lines to your synthesis script, after the: "**`set current_design`**", and before the: "**`compile -map_effort medium`**" lines.

2) Setting Input Delay

Input delay needs to be specified for all inputs (except the clock). In our case we will use the same timing for all inputs. This would not be the case in a "real" design. To aid us in setting a uniform input delay across all inputs we will setup a variable "**`prim_inputs`**" that holds the name of all our input ports except clock.

```
dc_shell> set prim_inputs [remove_from_collection [all_inputs]\
          [find port clk]]
```

Now prim_inputs is now the name of a variable that holds the name of all input ports except the clock. We can use this variable in our input delay statement.

```
dc_shell> set_input_delay -clock clk 0.5 $prim_inputs
```

Add the above two lines to your synthesis script.

3) Set drive strength of inputs

Synopsys needs to know how strongly driven inputs are so it can calculate the slope of the rise/fall transitions. This is necessary to get accurate timing assessment. There are two commonly used commands to set drive strength. One command "`set_driving_cell`" tells Synopsys that the input is driven with the same strength as if it was driven by a specified cell in the cell library. The other method "`set_drive`" just specifies a drive strength in ohms of the driver of the input.

The "`set_driving_cell`" method will be demonstrated for all inputs. We will the use the set_drive command to specify the drive strength for reset separately.

```
dc_shell> set_driving_cell -lib_cell NAND2X1_RVT -library\
          saed32rvt_tt0p85v25c $prim_inputs
```

This is telling Synopsys that all inputs specified in $prim_inputs are driven as strongly as if they were driven by a size 1 2-input NAND gate from our saed32 library.

Reset is another special signal like clock, and also has a very high fanout. We will use a **`set_drive`** command to tell Synopsys that *`rst_n`* is very strongly driven, so it does not think it needs to buffer it.

NOTE: High fanout signals will arrive to blocks at different times. A buffer would slow it down for "closer blocks" so everything arrives at approximately the right time. However, rst_n usually comes from an external chip, so it is already strongly driven and "fast". We don't want the synthesis tool trying to alter it as its not needed.

```
dc_shell> set_drive 0.0001 rst_n   ## Default unit for Ohms is MΩ
```

Add the two lines above to your synthesis script.

4) Setting Output Delay Constraints

For all outputs we need to tell Synopsys two things:

1.) When the output needs to be valid.
2.) How much capacitive load the output has to drive.

```
dc_shell> set_output_delay –clock clk 0.5 [all_outputs]
```

With the above line we are telling Synopsys that we need all outputs valid 0.5ns prior to the next rising edge of clock.

Now we will tell Synopsys how much load the output has to drive. The more load an output has to drive the slower it will be. Synopsys has to know that it needs to upsize the output driver (i.e use a stronger gate) to deal with the large capacitive load.

```
dc_shell> set_load 60 [all_outputs]  ## default unit for cap is fF
```

This command informs Synopsys that all outputs have to drive a 60fF load.

Add the two commands above to your synthesis script.

5) Some Miscellaneous Constraints

For internal nodes in the design there are parasitic routing capacitances (arise from the geometry of the circuit). These parasitic routing capacitances can adversely affect timing (capacitance slows down voltage transitions). We need to have Synopsys at least make a guess as to what these parasitic capacitances might be so it can design the logic to be fast enough to overcome them.

```
dc_shell> set_wire_load_model –name 16000 \
          –library saed32rvt_tt0p85v25c
```

We discuss parasitic capacitance and wireload models a bit later in the class. For now just know that the above line is setting up a heuristic way for it to estimate the capacitance of the wiring between gates. Wireload models vary with the size of the design. I larger design will have longer interconnects. We are saying use a wireload model for a block that is roughly 16000 square microns in size.

An internal design rule that Synopsys has to make regards the transition time of nodes. If nodes are too slow to transition they will spend too much time in a region with both high Vds and high Vgs. This can damage the transistors and slow them down over time. We inform Synopsys that all nodes in the design must have transition times of 0.1ns or faster.

```
dc_shell> set_max_transition 0.1 [current_design]
```

Add the above two lines to your synthesis script.

That covers the common timing and design rule constraints. The middle of your synthesis script should look something like:

```
##############################
# Constrain and assign clock #
##############################
create_clock -name "clk" -period 2 {clk}
set_dont_touch_network [find port clk]

##############################################
# Constrain input timings and Drive strength #
##############################################
set prim_inputs [remove_from_collection [all_inputs] [find port clk]]
set_input_delay -clock clk 0.5 $prim_inputs
set_driving_cell -lib_cell NAND2X1_LVT -library\
                     saed32lvt_tt0p85v25c $prim_inputs
set_drive 0.0001 rst_n

#####################################
# Constrain output timings and load #
#####################################
set_output_delay -clock clk 0.5 [all_outputs]
set_load 60 [all_outputs]

##################################
# Set wireload & transition time #
##################################
set_wire_load_model -name 16000 -library saed32lvt_tt0p85v25c
set_max_transition 0.1 [current_design]
```

NOTE: these should be RVT (not LVT). This example was from a semester where we used the low V_T version of the process

Figure 5-1 : Input Delay

Lesson 6 – Analyzing Results (reports)

Now start with a fresh design_vision session and execute your new synthesis script. One thing you may notice is the synthesized area is now larger that it was the last time. This is because the design has constaints (both timing and design rule) and Synopsys has to work harder and use more gates to meet those constraints.

How do we know our result is good. One handy command to execute after synthesis completes is "**`check_design`**".

After running the synthesis script you should execute the **`check_design`** command. You may get a warning about a signal *Bp* which is an input, but is also driven by a tri-state driver. This is expected for our design, however the output of **`check_design`** is a good place to look to make sure there are no strange unexpected topologies that are part of your resulting circuit.

1) Max Delays (Is the result fast enough?)

One of the main things to check is did your design meet timing from a max delay point of view. Meaning, is it fast enough? Does it meet setup times of internal flops, and output delay constraints of primary outputs. To check this we use the "**`report_timing –delay max`**" command.

`dc_shell> report_timing –delay max`

The report that results from this command shows that we are missing timing. The problem seems to end at port *Bp*. There is a large delay (1.33ns on the output driver TNBUFFX8 in my report)

We will look into fixing this later.

2) Min Delays (Is the result too fast…or do we have too much clock skew)

Meeting hold time of our flops is often more important than making setup times. If the design is too slow we can slow down the clock, and still test the chip. If we have a hold time problem we are in trouble at any clock speed. To check for hold time problems we use the "**`report_timing –delay min`**" command.

```
dc_shell> report_timing –delay min
```

The results from this report show we are meeting timing. The clock to Q timing of the flops is greater than the hold time requirements of the flops.

3) How Big is the Design?

To determine the size of your design use the "report_area" command. This will tell you the size of the design in square microns. It also gives reports about the number of cells used, and a breakdown of whether the cells are combinational or sequential. If the wireload model is complete it will also give an estimate of routing area.

```
dc_shell> report_area
```

Another handy thing one can do with reports is to re-direct the output to a file. This way you have a permanent record of your reports. This applies to timing reports as well as area reports. An example with an area report is shown next.

```
dc_shell> report_area > area.rpt
```

Add lines for timing and area reports to your script. These lines should be added after the "compile –map_effort medium" command.

Lesson 7 – Further Constraints and Optimizations

There are other constraints that have not been covered so far. On big one is clock skew. Modern digital designs will have huge numbers of flops. The clock cannot be distributed to all these flops uniformly (we try, but it is impossible). Therefore, one must specify a clock uncertainty so that Synopsys can ensure min delay timings (hold times) even in the presence of an imperfect clock. I can never remember what the command is, but I know it has the word "**`uncertainty`**" in it. Type the following commands in the dc_shell command line, not in the script yet.

```
dc_shell> help *uncertainty*
```

This command returns a list of all Synopsys commands that have "**`uncertainty`**" as part of the command name. I see a command for "**`set_clock_uncertainty`**" and this makes me say: "Ohh yeah that was it"!

So now I need to know the details of the "set_clock_uncertainty" command. I can type:

```
dc_shell> set_clock_uncertainty –help
```

That returns a more detailed help of the command. Now I can see how I want to use it.

```
dc_shell> set_clock_uncertainty 0.15 clk
```

Now that we have applied a clock uncertainty what does our min delay timing look like? Lets do a report again.

```
dc_shell> report_timing -delay min
```

If you look at the report you can see there is now a line stating the clock uncertainty of 0.15ns, and now we are no longer making it (we have negative slack on min delay).

So how do we fix it? Well we informed Synopsys about the clock uncertainty after it had compiled the design, so all we need to do is compile again, and it should fix the hold times right?

```
dc_shell> compile -map_effort medium

dc_shell> report_timing -delay min
```

What? It didn't fix it. Even after compiling again Synopsys didn't fix our hold time problem? Yes, it is bizarre, but we specifically have to request Synopsys to fix hold time problems. This is off by default.

```
dc_shell> set_fix_hold clk

dc_shell> compile -map_effort medium

dc_shell> report_timing -delay min
```

Good! Now it is fixed. Add the following lines to your synthesis script. Add them after the "**compile -map_effort medium**", but before the timing and area reports.

```
set_clock_uncertainty 0.15 clk
set_fix_hold clk
compile -map_effort medium    # 2nd compile
```

Our design is looking pretty good now, except for the issue with max delay timing. We have this strange problem with the ***Bp*** input/output having this large delay at the output. This stems from the fact that ***Bp*** is driven by a tri-state and the largest tri-state in this library can't drive a 60fF load with decent timing. We will reduce the output load specified on both ports driven by tri-states (***Bp*** & ***Am***)

```
dc_shell> set_load 30 [find port Bp]
dc_shell> set_load 30 [find port Am]
```

We also want to flatten the design so the resulting netlist written out has no hierarchy and is just instances of cells from the saed32 library.

```
Dc_shell> ungroup -all -flatten
```

Now write your synthesis script to disk. The middle/end part should look like:

```
#######################################
# Constrain output timings and load #
#######################################
set_output_delay -clock clk 0.5 [all_outputs]
set_load 60 [all_outputs]
set_load 30 [find port Bp]
set_load 30 [find port Am]

####################################
# Set wireload & transition time #
####################################
set_wire_load_model -name 16000 -library saed32lvt_tt0p85v25c
set_max_transition 0.1 [current_design]

##########################
# Now kick off synthesis #
##########################
compile -map_effort high
```

Again…this should be rvt

Remove the entire design from the synthesis tool, and run the synthesis script from scratch. Check out the results.

```
dc_shell> remove_design -all
dc_shell> source commMod.dc
```

DC is a large CAD tool that has been around for over 30 years. There are many more commands available to constrain and optimize. Pay attention in class, and we will cover a few more tricks. This tutorial, however, should be enough to get you on your way.